The Funding Window: Predicting VC Investment and Startup Outcomes Through Market-Aware Machine Learning

Fawwaaz Sheik, Shariq Abdul Aziz, Vedanth Chintala • CS439 • Rutgers University • 5/10/26
GitHub: *https://github.com/ved567/CS439_FinalProject*

Abstract

Venture capital funding decisions are traditionally studied through startup-level characteristics such as founding team quality, sector, and traction metrics. However, funding outcomes are also shaped by macroeconomic conditions including interest rate regimes, market volatility, and sector-level attention cycles. In this paper, we present The Funding Window, a macro-aware machine learning framework that predicts whether a startup will be acquired by fusing 31 startup-level features with three contemporaneous macroeconomic signals. We benchmark an XGBoost model against a logistic regression baseline, achieving a ROC-AUC of 0.802 versus 0.784, and demonstrate via SHAP analysis that all three macro signals such as Federal Funds Rate, VIX, and Google Trends sector interest all rank within the top 15 of 34 features by predictive contribution. Our results confirm that external market conditions contribute meaningfully to acquisition outcomes independent of startup-intrinsic factors, with network connectivity emerging as the single strongest individual predictor.

---

1 Introduction

The question of which startups achieve successful acquisition has significant economic and social implications. Venture capital deployment reached record highs in 2021 before contracting sharply in 2022 following Federal Reserve rate hikes, yet most predictive models treat market conditions as static background noise rather than dynamic, per-observation inputs. A startup pitching in a low-interest-rate, high-liquidity environment faces fundamentally different odds than an identical company pitching during a credit tightening cycle.

Existing tools for startup outcome prediction are built primarily for institutional investors, operate as black boxes, and neglect the macroeconomic context in which funding decisions are made. No public model explicitly captures the interaction between a startup's intrinsic qualities and the prevailing economic climate at the moment of pitch. This paper makes the following contributions:

- A macro-aware feature set combining 31 startup attributes with Federal Funds Rate, VIX volatility index, and Google Trends sector interest, each snapshotted at the startup's first funding date to prevent look-ahead bias.

- A time-based train/test split (pre-2010 training, 2010-2013 test) that simulates real-world deployment and avoids data leakage.
- A baseline comparison of logistic regression versus XGBoost achieving ROC-AUC of 0.784 and 0.802 respectively on the held-out test set.
- SHAP-based explainability shows that all three macro signals rank among the top 15 of 34 predictive features, confirming market conditions contribute independently of startup quality.

---

2 Related Work

2.1 Startup Success and Funding Prediction

Prior work on startup outcome prediction has primarily relied on structured Crunchbase data. Zbikowski & Antosiuk (2021) demonstrated that logistic regression, SVM, and gradient boosting can predict funding outcomes using VC and angel investor indicators, funding rounds, and geographic features. Gangwani et al. (2023) extended this by engineering features around the triangular relationship between investors, startups, and markets. These works establish a strong supervised learning baseline but treat market conditions as static background rather than dynamic per-observation inputs which is the gap our work directly addresses.

2.2 Macroeconomic Signals in Capital Markets

Macroeconomic variables such as interest rates and market volatility indices have been widely shown to predict aggregate capital flows. Gompers & Lerner (2000) demonstrated that committed capital to VC funds correlates strongly with economic cycles and capital gains tax rates. Kenney & Zysman (2019) documented how the post-2008 low-rate environment structurally expanded VC deployment. However, to our knowledge no prior work incorporates these signals at the individual startup level, temporally aligned to each company's specific funding attempt.

2.3 Explainability in Financial ML

SHAP (Shapley Additive Explanations), introduced by Lundberg & Lee (2017), has become the standard method for feature attribution in gradient boosted models. Applications in credit scoring and financial risk modeling have used SHAP to decompose predictions into interpretable per-feature contributions. We apply this framework to startup prediction to isolate the macro signal contribution from startup-intrinsic factors.

---

3 Methodology

3.1 Data Sources

We draw from three data sources contributing distinct feature families. For startup data, we use a Crunchbase-derived dataset sourced from Kaggle containing 923 startups founded between 2000 and 2013, with features spanning funding history, geography, industry category, investor type, and milestone activity. The binary target encodes acquisition (1) versus closure (0). For macroeconomic signals, we retrieve Federal Funds Rate (FEDFUNDS) and CBOE VIX index (VIXCLS) from the FRED API at monthly granularity and join each startup by funding month. For sector trends, we use Google Trends normalized search interest scores for each startup's primary industry keyword at the time of first funding.

3.2 Preprocessing and Feature Engineering

Identifier columns were dropped. The target was binarized yielding a 65/35 class distribution. Forty-six records with negative age-to-first-funding values which were attributable to Crunchbase data entry inconsistencies. These were clipped to zero. The funding_total_usd column exhibited extreme right skew (max $5.7B versus mean $25M) and was log-transformed. Missing milestone age values (152 rows) were imputed using per-category medians, preserving sector-level distributional differences. Missing sector_trend values (44 rows) were imputed using per-category means. All continuous features were standardized using StandardScaler fit exclusively on the training set to prevent leakage.

3.3 Temporal Split Strategy

A time-based split in 2010 yielded 709 training observations and 214 test observations (77/23 split). Random shuffling was deliberately avoided: using future observations during training constitutes data leakage and overestimates real-world performance. The temporal split also ensures the test set reflects a distinct macroeconomic regime, the post-financial-crisis recovery, providing a more demanding evaluation scenario.

3.4 Models

Two models are compared. The baseline is an L2-regularized logistic regression with class_weight='balanced' for imbalance handling and max_iter=1000. The primary model is XGBoost with scale_pos_weight=0.55 (326/597) for imbalance and hyperparameters tuned via 5-fold cross-validation optimizing ROC-AUC on the training set. Final hyperparameters: n_estimators=200, max_depth=4, learning_rate=0.1, subsample=0.8.

---

4 Experiments and Results

4.1 Model Performance

Table 1 reports performance of both models on the held-out test set. XGBoost outperforms logistic regression on ROC-AUC (0.802 vs 0.784) and F1 (0.720 vs 0.711). Accuracy is marginally lower for XGBoost (0.692 vs 0.696), illustrating why accuracy is misleading forimbalanced classification: XGBoost trades easy majority-class predictions for better overall discrimination, reflected in its superior AUC.

Table 1: Test-Set Performance Comparison

Metric	Logistic Regression	XGBoost (Champion)
ROC-AUC	0.784	**0.802 ★**
F1-Score	0.711	**0.720 ★**
Accuracy	0.696	0.692

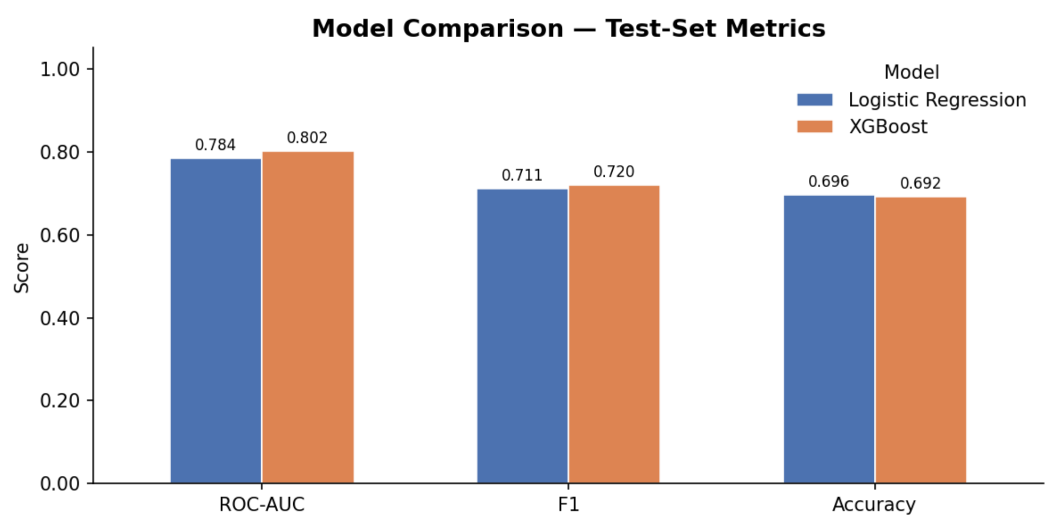

Figure 1: Bar chart comparison of ROC-AUC, F1, and Accuracy. XGBoost leads on the two primary metrics; accuracy diverges, illustrating its unsuitability as sole metric for imbalanced classification.

4.2 Confusion Matrix Analysis

Figure 2 presents side-by-side confusion matrices for both models on the 214-observation test set. Logistic regression correctly identifies 69 closed startups (true negatives) versus XGBoost's 63, while XGBoost correctly identifies 85 acquired startups (true positives) versus 80 for logistic regression. This tradeoff reflects differing calibration strategies: logistic regression is more conservative in flagging closures, while XGBoost is more sensitive in detecting acquisitions which we believe is the more commercially relevant property for a founder-facing application.

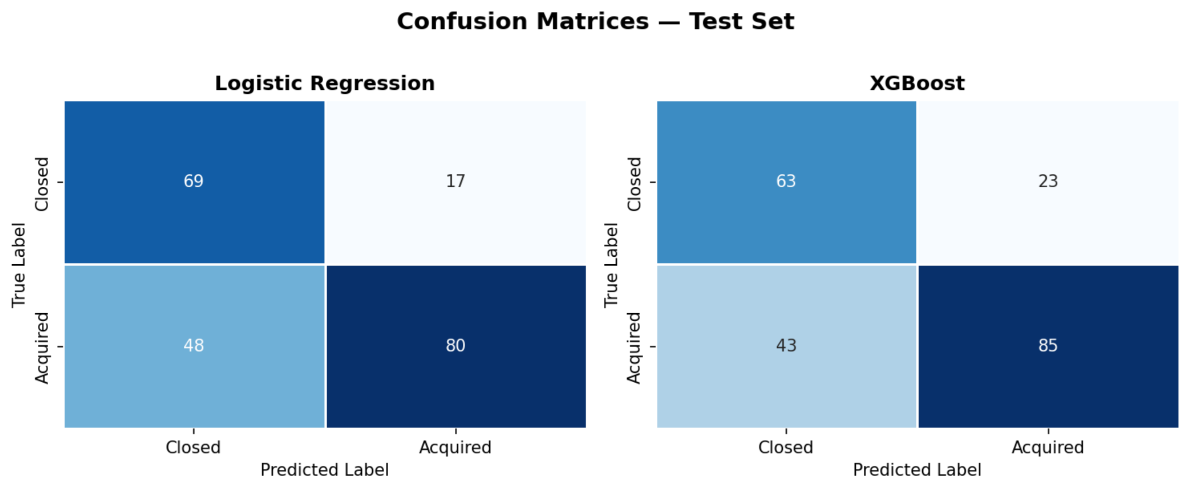

Figure 2: Confusion matrices for Logistic Regression (left) and XGBoost (right). XGBoost identifies more true acquisitions (85 vs 80) at the cost of slightly more false positives among closed startups.

4.3 PCA Feature Space Visualization

Figure 3 presents a PCA projection of the 214 test-set observations onto two principal components. PC1 explains 26.1% of variance and PC2 explains 15.7%, together capturing 41.8% of the 34-dimensional feature space. Closed startups cluster in the upper-left quadrant while acquired startups dominate the center and right regions. The substantial overlap is consistent with the difficulty of the prediction task and the known noisiness of Crunchbase outcome labels, but the directional separation validates that our feature set encodes a genuine discriminative signal.

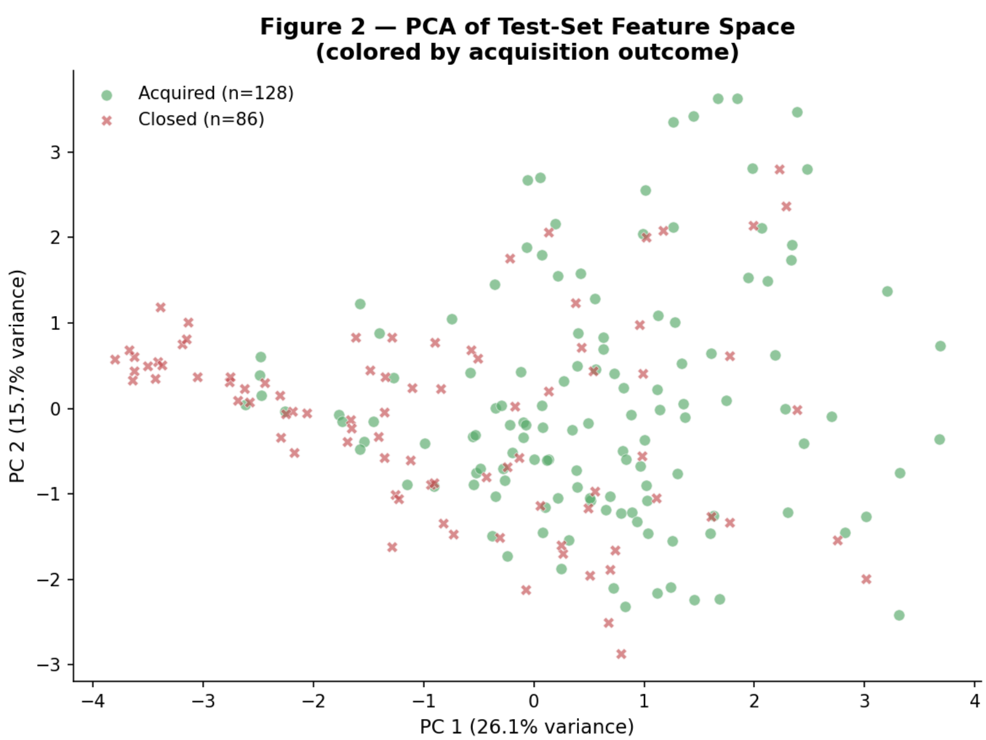

Figure 3: PCA projection of test-set startups. Closed startups (red crosses) cluster upper-left; acquired startups (green circles) dominate center and right. Two components explain 41.8% of total variance.

4.4 SHAP Feature Importance

Figures 4 and 5 present SHAP analysis for the XGBoost model. The core finding: all three macro signals rank within the top 15 of 34 features — fed_rate at rank 12, sector_trend at rank 13, vix at rank 14 — confirming our thesis that market conditions contribute independently of startup quality.

The dominant feature is relationships (mean |SHAP| = 0.68), a proxy for founder network connectivity, nearly 2.5x the next feature. The second tier: funding_total_usd (0.26), age_last_milestone_year (0.23), milestones (0.22), and avg_participants (0.21) captures operational traction. Macro signals contribute above the long tail of geographic and sector binary flags. The beeswarm plot reveals that high fed_rate values exert a modest negative effect on acquisition probability, consistent with the hypothesis that monetary tightening suppresses VC activity at the individual deal level.

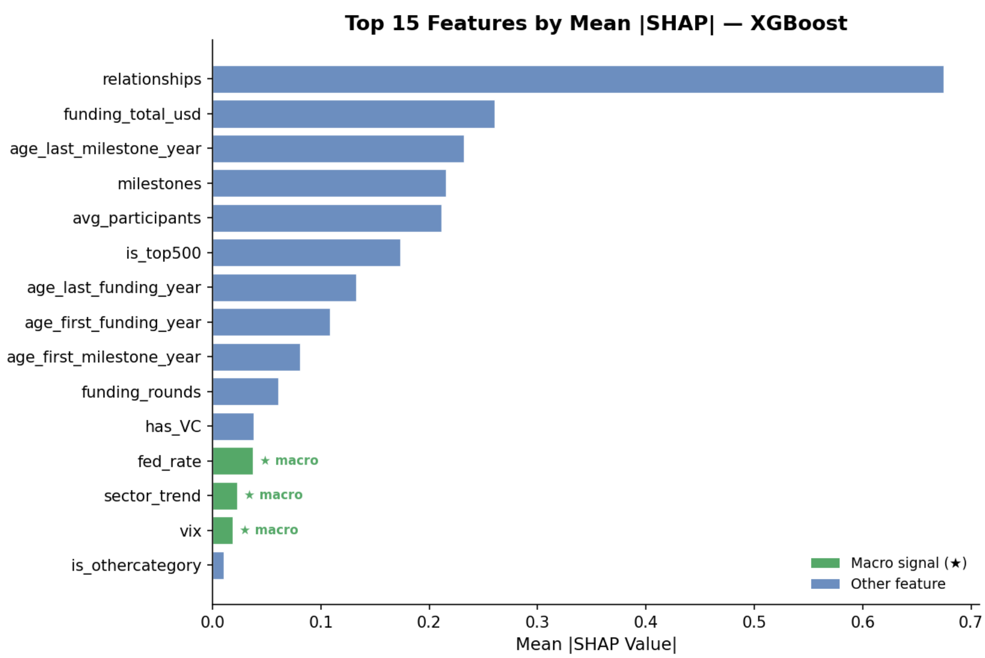

Figure 4: Top 15 features by mean |SHAP| value. Macro signals (green, starred) rank 12-14 of 34, above all geographic and most sector features. Relationships dominates at 0.68.

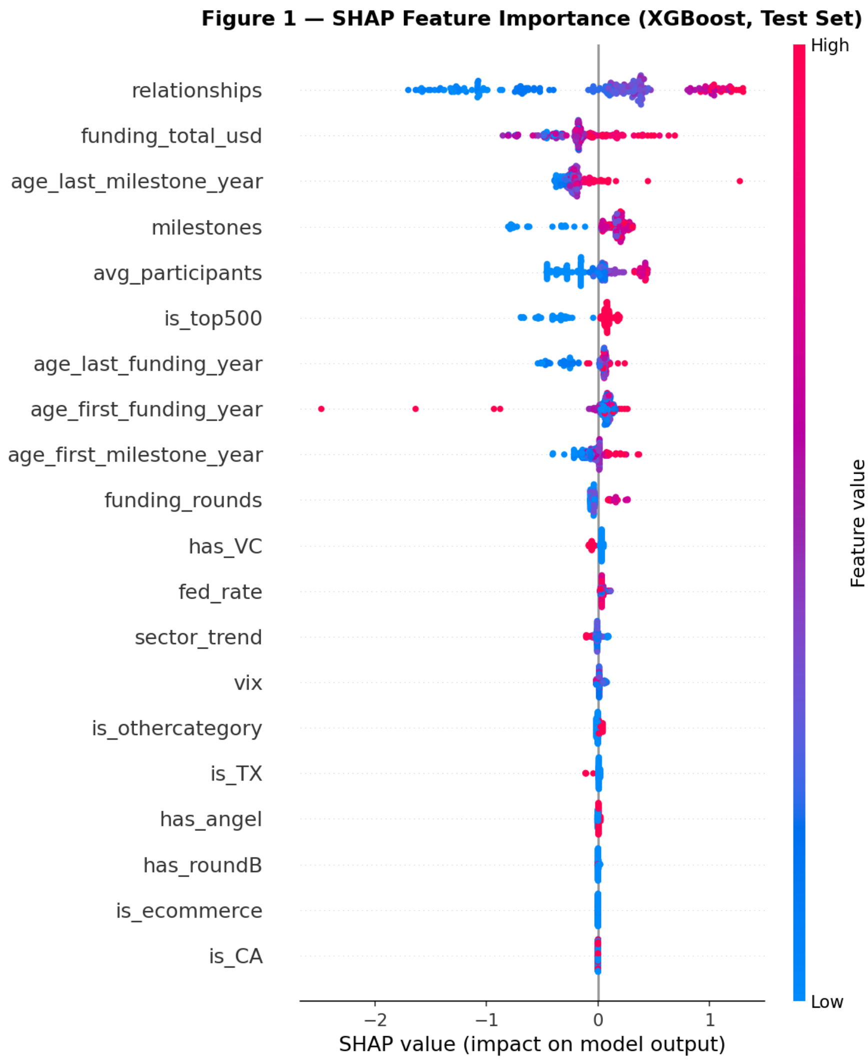

Figure 5: SHAP beeswarm plot. Each point is one test observation; color indicates feature value (pink=high, blue=low). Relationships SHAP values span approximately -2.5 to +1.5, indicating its strong bidirectional influence on predictions.

---

5 Conclusion

We presented The Funding Window, a macro-aware framework for predicting startup acquisition outcomes. Our XGBoost model achieves ROC-AUC of 0.802 on a temporally held-out test set, outperforming a logistic regression baseline (0.784). SHAP analysis confirms that Federal Funds Rate, VIX, and sector-level Google Trends interest contribute independently and meaningfully alongside traditional startup metrics. The dominant role of network connectivity raises an important question: do funding outcomes reflect startup quality, or the social capital of founders? The presence of macro signals in the top 15 features suggests a further confound; that market timing may account for a non-trivial share of outcomes attributed to startup quality in models that ignore economic context.

5.1 Limitations

- Survivorship bias: Crunchbase overrepresents startups that publicly sought funding, potentially inflating acquisition rates and underrepresenting stealth failures.
- Dataset recency: Coverage through 2013 predates the 2021 VC boom and 2022 correction; replication on a more recent cohort would strengthen generalizability.
- Small test set: The temporal split yields 214 test observations, limiting statistical precision of evaluation metrics.
- Macro signal granularity: Monthly snapshots may miss within-month volatility; VC-specific metrics such as dry powder levels could add precision.

5.2 Future Work

- Train separate models per market era to directly measure how macro regime modulates feature importance over time.
- Build a founder-facing tool that produces fundability scores with counterfactual explanations identifying which features a founder can realistically change.
- Extend to more recent data and add NASDAQ composite and IPO market activity as additional macro signals.

References

Gompers, P., & Lerner, J. (2000). Money chasing deals? The impact of fund inflows on private equity valuations. Journal of Financial Economics, 55(2), 281-325.

Kenney, M., & Zysman, J. (2019). Unicorns, Cheshire cats, and the new dilemmas of entrepreneurial finance. Venture Capital, 21(1), 35-50.

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30.

Zbikowski, K., & Antosiuk, P. (2021). A machine learning, bias-free approach for predicting business success using Crunchbase data. Information Processing & Management, 58, 102555.

Gangwani, P., et al. (2023). Predicting startup success using triangular investor-business-market relationships. Journal of Business Venturing Insights, 19, e00371.